

ResolveNow: Online Complaint Registration and Management System — Project Report

1. Introduction

This project, **ResolveNow**, is a full-stack online complaint registration and management system developed using the **MERN** (MongoDB, Express.js, React.js, Node.js) stack. It enables individuals or organizations to securely submit, track, and resolve complaints, providing a centralized platform that optimizes complaint handling, ensures regulatory compliance, and improves customer satisfaction.

2. Purpose

ResolveNow empowers users to file complaints easily and track their resolution in real-time. It helps organizations streamline complaint workflows, ensure data security, and facilitate smooth communication between users, agents, and admins.

3. Features

ResolveNow includes the following core features:

User Registration and Login:

Users can sign up, verify accounts, and securely log in.

Complaint Submission:

Users submit complaints with detailed descriptions, attachments, and contact info.

Tracking & Notifications:

Users receive real-time updates about their complaint status via dashboard and email.

Interaction with Agents:

Built-in messaging allows direct chat between users and agents for faster resolution.

Assigning & Routing Complaints:

Admin intelligently assigns complaints to the right agent or department.

Security & Confidentiality:

Authentication, data encryption, and strict access control protect user data.

4. How I Did It: Technical Implementation

My approach followed a structured development process across four key milestones, utilizing the MERN stack for both frontend and backend development.

4.1. Technical Architecture

ResolveNow follows a **client-server architecture**:

1. **Frontend: React.js with Bootstrap and Material UI** for an intuitive UI. **Axios** is used to call **RESTful APIs**.
2. **Backend: Node.js with Express.js** handles server-side logic and **APIs**.
3. **Database: MongoDB** stores user data, complaints, agent assignments, and messages.
4. **Real-Time Communication: Socket.io or WebRTC APIs** can be used for live agent-user interaction if needed.

4.2. Database Design (ER Diagram)

Our MongoDB database includes:

- **User Schema:** Stores user details (name, email, password, userType).
- **Complaint Schema:** Stores complaint details (userId, name, description, address, status).
- **Assigned Complaint Schema:** Links complaints to agents.
- **Message Schema:** Manages chat between users and agents linked by complaintId.

4.3. Project Flow (User Journey)

User Flow:

1. Register & Login
2. Submit Complaint
3. Track Complaint Status & Receive Notifications
4. Interact with Assigned Agent
5. Receive Resolution & Give Feedback

Agent Flow:

1. Register & Login
2. View Assigned Complaints
3. Communicate with Users
4. Update Complaint Status

Admin Flow:

1. Monitor all complaints
2. Assign complaints to agents
3. Manage users and agents

4.4. Development Milestones

Milestone 1: Project Setup & Configuration

- Installed Node.js, npm, MongoDB, React.js.
- Created separate **client** and **server** folders.
- Installed packages: Express, Mongoose, Axios, Bootstrap, Material UI, Socket.io.

Milestone 2: Backend Development

- Set up Express server and middleware.
- Created authentication, complaints, and chat APIs.
- Defined Mongoose schemas for users, complaints, agents, messages.
- Implemented JWT authentication for secure access.

Milestone 3: Database

- Connected MongoDB via Mongoose.
- Created and tested all schemas & relationships.

Milestone 4: Frontend Development

- Built UI using React.js with reusable components.
- Implemented routing and API integration.
- Designed pages: Login, Register, Dashboard, Complaint Form, Complaint Status, Agent & Admin dashboards.

Milestone 5: Implementation & Testing

- Tested user flows, complaint submission, agent interaction, admin workflows.
- Verified real-time updates and notifications.

5. Technologies Used

- **Frontend:** React.js, Bootstrap, Material UI, Axios, HTML, CSS, JavaScript
- **Backend:** Node.js, Express.js, Socket.io/WebRTC API
- **Database:** MongoDB, Mongoose
- **Version Control:** Git, GitHub
- **IDE:** Visual Studio Code

6. Conclusion

ResolveNow successfully demonstrates a complete MERN-based online complaint registration and management system. It provides a secure, user-friendly, and efficient platform to handle complaints, track progress, communicate with agents, and maintain high user satisfaction through modern web technology.